

Thermo

HW #3

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a.

- ① For water. $P = 100 \text{ psi}$, $u = 500 \text{ Btu/lb}$, determine $T(^{\circ}\text{F})$, $v(\text{ft}^3/\text{lb})$, and $h(\text{Btu/lb})$.

Step 1 calculate quality of steam @ state 1

$$u_1 = u_f + x(u_g - u_f)$$

$$500 = 298.19 + x(1105.5 - 298.19)$$

$$x = 0.25$$

Step 2, enthalpy @ state 1

$$h = h_f + x(h_g - h_f)$$

$$h = 298.51 + 0.25(1187.5 - 298.51)$$

$$h = 520.75 \text{ Btu/lb}$$

Step 3, specific vol (ft^3/lb)

$$v = v_f + x(v_g - v_f)$$

$$v = .01774 + .25(4.4321 - .01774)$$

$$v = 1.121 \text{ ft}^3/\text{lb}$$

Temp @ state 1

$$T = T_{\text{sat}@100 \text{ psi}} = 327.81^{\circ}\text{F}$$

$$T = 626.3^{\circ}\text{F}$$

$$T = T_{\text{sat}@100 \text{ psi}} = 327.81^{\circ}\text{F}$$

$$62.6 - 300 = 221.81$$

$$\frac{T_1 - T_2}{T_1 - T_x} = \frac{P_1 - P_2}{P_1 - P_x}$$

$$\frac{330 - 320}{330 - T_x} = \frac{103 - 89.6}{103 - 500}$$

$$\frac{10}{330 - T_x} = \frac{13.4}{-397}$$

$$-3970 = 4422 - 139T$$

$$-8392 = -139T$$

$$T = 626.3^{\circ}\text{F}$$

b. For ammonia $T = 40^{\circ}\text{F}$, $x = 0.5$,

determine P in lb/in^2 , $v(\text{ft}^3/\text{lb})$, and $u(\text{Btu/lb})$

$$u_f = 85.98 \text{ Btu/lb}$$

$$u_g = 568.42 \text{ Btu/lb}$$

$$v_f = .02533 \text{ ft}^3/\text{lb}$$

$$v_g = 3.9669 \text{ ft}^3/\text{lb}$$

$$x = .50$$

$$u = u_f + x(u_g - u_f)$$

$$u = 85.98 + .5(568.42 - 85.98)$$

$$u = u_f + x(u_g - u_f)$$

$$u = 85.98 + .5(482.94)$$

$$u = 327.2 \text{ Btu/lb}$$

$$v = .02533 + .5(3.94)$$

$$v = 2 \text{ ft}^3/\text{lb}$$

$$P = 73.359 \text{ lb}/\text{in}^2$$

c. For water $P = 80 \text{ bar}$, $v = 0.04034 \text{ m}^3/\text{kg}$, T in $^\circ\text{C}$; u in kJ/kg ; h in kJ/kg ;
Super heated region

From the table:

$$\begin{bmatrix} T_3 = 480^\circ\text{C} \\ u_3 = 3025.7 \text{ kJ}/\text{kg} \\ h_3 = 3348.4 \text{ kJ}/\text{kg} \end{bmatrix}$$

d. For refrigerant 134a @ $T = -20^\circ\text{C}$ and $h = 235.31 \text{ kJ}/\text{kg}$
 P (bar), v (m^3/kg), u (kJ/kg)?

Saturated vapor curve.

$$\begin{bmatrix} P = 1.3249 \text{ bar} \\ v = 0.1464 \text{ m}^3/\text{kg} \\ u = 215.84 \text{ kJ}/\text{kg} \end{bmatrix}$$

② Rigid Tank constant volume. initially saturated vapor @ 180°C
is cooled to 100°C

a. initial P (MPa)

b. final quality (x)

$T_1 = 180^\circ\text{C}$ $T_2 = 100^\circ\text{C}$, From table, sat vapor @ 180°C

$$v_1 = v_2$$

$$P_1 = 1.022 \text{ MPa}, v_1 = 0.19405 \text{ m}^3/\text{kg}$$

@ 100°C $v_{f2} = 1.0435 \times 10^{-3} \text{ m}^3/\text{kg}$

$$v_{g2} = 1.673 \text{ m}^3/\text{kg}$$

$$v_{f2} < v_2 < v_{g2}$$

So, @ 100°C $P_2 = 1.014 \text{ bar} = 1.014 \text{ MPa}$

b. quality. $v_2 = v_{f2} + x_2(v_{g2} - v_{f2})$

$$x = \frac{0.19405 - (1.0435 \times 10^{-3})}{1.673 - (1.0435 \times 10^{-3})} = 0.1154$$

$$x = 0.1154$$

HW 3

- ③ 2 kg water vapor in piston expands @ constant pressure of 300 kPa (3.0 bar) from a saturated vapor state to a $V_2 = 2.064 \text{ m}^3$

a. initial temp? b. final temp c. work (kJ)

given

$$m = 2 \text{ kg}$$

from table @ $P = 300 \text{ bar}$

$$P_1 = 3.0 \text{ bar} = P_2$$

$$v_g = 0.6055 \text{ m}^3/\text{kg} \times 2 \text{ kg} = 1.211 \text{ m}^3$$

$$V_2 = 2.064 \text{ m}^3$$

$$V_1 = 1.211 \text{ m}^3$$

$$\text{C. } W = -\int P dv = -P(V_2 - V_1)$$

$$W = 300(2.064 - 1.211) = 255.9 \text{ kJ} \times \frac{1000 \text{ J}}{1 \text{ kJ}}$$

$$W = 0.2559 \text{ kJ}$$

$$\text{a. initial temp } T_1 = 133.6^\circ\text{C}$$

$$\frac{V_1 P_1}{T_1} = \frac{V_2 P_2}{T_2} \Rightarrow \frac{1.211(300)}{T_1} = \frac{2.064(300)}{T_2}$$

From sat L-v table

$$T_1 @ 3.00 \text{ bar}$$

$$= 133.6^\circ\text{C}$$

$$2.064 T_1 = 1.211 T_2$$

$$2.064(133.6)$$

$$\frac{1.211}{1.211} = T_2$$

$$\text{b. } T_2 = 227.7^\circ\text{C}$$

$$\text{④ } P_1 = 10 \text{ bar}$$

$$P_2 = 2 \text{ bar}$$

$$T_1 = 60^\circ\text{C}$$

$$T_2 = 30^\circ\text{C}$$

superheated

or saturated?

$$\text{mass} = 1 \text{ kg}$$

$$W \text{ done by refrigerant} = 20 \text{ kJ}$$

$$\text{a. } V_1 \text{ in m}^3 \quad \text{b. } Q \text{ in kJ}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

a. From saturated liquid vapor table

$$v_g = 0.01434 \text{ m}^3/\text{kg} \times 1 \text{ kg} = 0.01434 \text{ m}^3$$

$$V_1 = 0.01434 \text{ m}^3$$

$$W = \Delta A - Q \quad \Delta U =$$

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b. $Q = u + W$

$$Q = (1 \text{ kg})(1.112) + 20 \text{ kJ} \quad u = mc\Delta T$$

$$Q = (1 \text{ kg})(1.112)(-30) + 20 \text{ kJ}$$

$$Q = -13.36 \text{ kJ}$$

⑤ closed, rigid tank liquid-vapor phase

$$m = 1.50 \text{ kg } H_2O$$

$$T_1 = 70^\circ\text{C}$$

heat transfer occurs to sat. vapor @ $T_2 = 120^\circ\text{C}$

a. $P_1 = ? \text{ kPa}$ b. $Q = ? \text{ kJ}$

↑

interpolation.

$$\frac{T_1 - T_2}{T_1 - T_x} = \frac{P_1 - P_2}{P_1 - P_x} \Rightarrow \frac{75.87 - 69.10}{75.87 - 70} = \frac{.40 - .30}{.40 - P_x}$$

$$1.153 = \frac{.12}{.4 - P_x}$$

$$\Rightarrow .12 = 0.4612 - 1.153P_x$$

$$P_x = 0.296 \text{ kPa}$$

60%

→ b. $Q = W + \Delta u$

?

⑥ 2 phase l-v Mixture of Refrigerant 22 @ 10 bar.

$m = 25 \text{ kg}$ $x = 60\% = 0.60$

a. Vol (m^3)

b. frac of total vol occupied by sat vapor

$v_f = 0.8352 \times 10^{-3} \text{ m}^3/\text{kg}$ $v_g = 0.0236 \text{ m}^3/\text{kg}$

$v = v_f + x(v_g - v_f) = 0.8352 \times 10^{-3} + 0.6(0.0236 - 0.8352 \times 10^{-3})$

$v = 0.01443 \text{ m}^3/\text{kg} \times 25 \text{ kg}$

~~$V = 0.36075 \text{ m}^3$~~

$x = \frac{m_g}{m_g + m_f} \Rightarrow 0.6 = \frac{m_g}{m_g + 25}$

$\star V_g = 0.0237 \times 37.5 = 0.885 \text{ m}^3$ $\therefore m_g = 37.5 \text{ kg}$

$V_f = 25(0.8352 \times 10^{-3}) = 0.02088 \text{ m}^3$

so vol of tank

$V = m v$

$V = (25 + 37.5)[0.01449] = 0.9056 \text{ m}^3$

b. $x_{\text{vap}} = \frac{V_g}{V} = \frac{0.885}{0.9056} \times 100 = 97.73\%$

⑦ 0.2 g helium

$T_1 = 27^\circ\text{C}$

$T_2 = 17^\circ\text{C}$

$P_1 = 1.3 \text{ bar}$

$P_2 = 0.80 \text{ bar}$

a. R ($\text{J/g}\cdot\text{K}$) b. V_1 (cm^3) c. V_2 (cm^3)

$M_{\text{He}} = 4.003 \text{ kg/kmol}$



⑦ Continued

$$R = \frac{8.314}{4.003} = 2.0785 \text{ kJ/kg} \cdot \text{K} \cdot \frac{1 \text{ kg}}{1000 \text{ g}} \cdot \frac{1 \text{ J}}{0.001 \text{ kJ}}$$

$$R = 2.0785 \text{ J/g} \cdot \text{K}$$

b. V_1 , $P_1 V_1 = m R_{HE} T_1$

$$V_1 = \frac{0.2 \times 10^{-3} \times 2.0785 \times 300}{1.3 \times 10^2} = 959 \text{ cm}^3$$

$$c. V_2 = \frac{0.2 \times 10^{-3} \times 2.0785 \times 290}{0.8 \times 10^2} = 1506 \text{ cm}^3$$

⑧ CO₂ closed rigid tank.

$$T_1 = 25^\circ \text{C}$$

$$T_2 = 257^\circ \text{C}$$

$$P_1 = 2.75 \text{ bar}$$

$$P_2 = P_1 = 2.75 \text{ bar}$$

$$V_1 = 0.05 \text{ m}^3$$

a. m_{CO_2} in kg, $P_1 V_1 = m R T_1$

$$20 \times 144 \times 2.2 = m \left[\frac{1545}{44.01} \right] \times (80 + 459.67)$$

$$m = 0.334 \text{ lbm} \times \frac{0.453}{1 \text{ lb}} \text{ kg} = 0.151 \text{ kg}$$

b. $W = Q - m(u_2 - u_1)$

From table, $u_1 = 2984 \text{ BTU/mol}$

$$u_2 = 6337.4 \text{ BTU/mol}$$

$$W = -2.6(0.3341 \times \frac{1}{40.1}) \times (6337.4 - 2984.4)$$

$$W = -30.56 \text{ BTU} \times \frac{1.055}{1 \text{ BTU}} = -32.24 \text{ kJ}$$